# Solving Equations

NC.8.EE.7 • Equations and inequalities in one variable

**Unit 3 • Front**

## 1 Inverse Operations: Do the Same Thing to BOTH Sides

**Adding?** Subtract

$$\begin{aligned} x+5 &= 12 \\ \underline{-5} &\quad \underline{-5} \\ x &= 7 \end{aligned}$$

**Subtracting?** Add

$$\begin{aligned} x-3 &= 8 \\ \underline{+3} &\quad \underline{+3} \\ x &= 11 \end{aligned}$$

**Multiplying?** Divide

$$\begin{aligned} 3x &= 15 \\ \frac{3x}{3} &= \frac{15}{3} \\ x &= 5 \end{aligned}$$

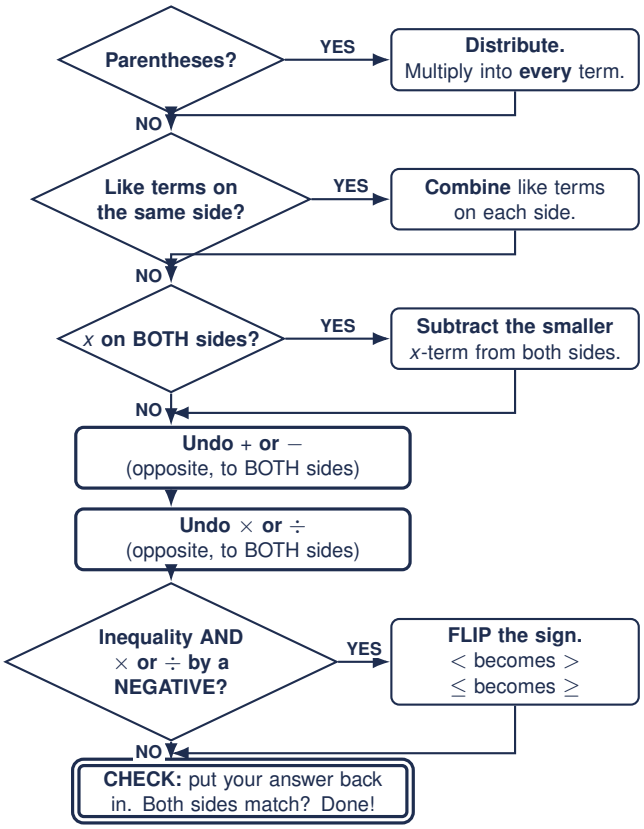
**Dividing?** Multiply

$$\begin{aligned} \frac{x}{4} &= 2 \\ 4 \cdot \frac{x}{4} &= 2 \cdot 4 \\ x &= 8 \end{aligned}$$

**Fraction?** Reciprocal

$$\begin{aligned} \frac{2}{3}x &= 6 \\ \frac{3}{2} \cdot \frac{2}{3}x &= 6 \cdot \frac{3}{2} \\ x &= 9 \end{aligned}$$

## 2 Follow the Flowchart (NO? Skip down to the next question.)




**Walk through it:** $-2x+1=8(x+2)-5x$

**Parentheses?** $-2x+1=8(x+2)-5x$
YES

Distribute 8 $-2x+1=8x+16-5x$

**Like terms?**
YES

$8x-5x=3x$ $-2x+1=3x+16$

**Both sides?**
YES

Smaller is $-2x$, so add $2x$ $\underline{+2x} \quad \underline{+2x}$

$1=5x+16$

**Undo +** $\underline{-16} \quad \underline{-16}$

$-15=5x$

**Undo ×** $\frac{-15}{5}=\frac{5x}{5}$

$-3=x$

**Inequality?** NO No flip. $\boxed{x=-3}$

**CHECK** $x=-3$:
Left: $-2(-3)+1=6+1=7$
Right: $8(-3+2)-5(-3)=-8+15=7$
$7=7$ ✓ Done!

## 3 Watch Out!

**Distribute to EVERY term.**
$3(x-4)=3x-12$ ✓
$3(x-4)\neq 3x-4$

**A negative goes in too.**
$-2(x+5)=-2x-10$ ✓
$-2(x+5)\neq -2x+10$

**Keep the sign with the term.**
$4-3x$ means $+4$ and $-3x$
Like terms: $5x$ and $-2x$

## 4 Translate Words to Math

| | |
|---|---|
| + | sum, more than, plus, increased by, total |
| − | difference, less than, minus, decreased by |
| × | product, times, of, twice, per |
| ÷ | quotient, divided by, split equally |
| = | is, equals, the same as, results in |

7 more than $n$ → $n+7$
5 less than $x$ → $x-5$ **(order flips!)**
twice a number → $2n$
a number split by 4 → $\frac{n}{4}$
3 times a number is 18 → $3n=18$

# Solutions & Inequalities

NC.8.EE.7 • Equations and inequalities in one variable



## 5 How Many Solutions?

**ONE solution**

$$\begin{aligned} 5x - 4 &= 2x + 8 \\ \underline{-2x} & \quad \underline{-2x} \\ 3x - 4 &= 8 \\ \underline{+4} & \quad \underline{+4} \\ 3x &= 12 \\ \frac{3x}{3} &= \frac{12}{3} \\ x &= 4 \end{aligned}$$

$x$ = one number

**NO solution**

$$\begin{aligned} 2(3x + 1) &= 6x + 5 \\ 6x + 2 &= 6x + 5 \\ \underline{-6x} & \quad \underline{-6x} \\ 2 &= 5 \end{aligned}$$

$x$'s cancel.
$2 = 5$ is **FALSE**.

No solution

**INFINITELY MANY**

$$\begin{aligned} 3(2x + 1) &= 6x + 3 \\ 6x + 3 &= 6x + 3 \\ \underline{-6x} & \quad \underline{-6x} \\ 3 &= 3 \end{aligned}$$

$x$'s cancel.
$3 = 3$ is **TRUE**.

Infinitely many solutions

**Shortcut:** Simplify each side. Then compare. $x$-terms **different** → one solution. $x$-terms **same**, numbers **different** → no solution. **Both sides identical** → infinitely many.

## 6 Inequalities: Solve Like an Equation, Then Graph

| Sign | Key Words | Circle |
|---|---|---|
| $<$ | less than, fewer than, below | open ○ |
| $>$ | more than, greater than, above | open ○ |
| $\leq$ | at most, no more than, maximum | closed ● |
| $\geq$ | at least, no less than, minimum | closed ● |

$x > 2$




$x \leq -1$




**Tip:** With $x$ on the left, the arrow points the same way as the sign.

**The FLIP rule:** $\times$ or $\div$ by a **negative** → flip the sign.

$$\begin{aligned} 6 - 2x &\leq 14 \\ \underline{-6} & \quad \underline{-6} \\ -2x &\leq 8 \\ \frac{-2x}{\mathbf{-2}} &\geq \frac{8}{\mathbf{-2}} \\ x &\geq -4 \end{aligned}$$

**Undo** $-$: subtract 6.

**Undo** $\times$: divide by $-2$. Negative! **FLIP** $\leq$ to $\geq$.

$x \geq -4$




**Check a point:** try $x = 0$. $6 - 2(0) = 6$. $6 \leq 14$ ✓

## 7 Word Problems

**1 Define** the variable. **2 Equation or inequality?** Look for the key words. **3 Write** it. **4 Solve** with the flowchart. **5 Answer** in a sentence. Does it make sense?

**Inequality:** A gym costs \$30 to join plus \$15 per month. You can spend **at most** \$120. How many months?

$m$ = number of months

$$\begin{aligned} 30 + 15m &\leq 120 \\ \underline{-30} & \quad \underline{-30} \\ 15m &\leq 90 \\ \frac{15m}{15} &\leq \frac{90}{15} \\ m &\leq 6 \end{aligned}$$

$$15 \overline{)\,90} = 6, \quad -90 = 0$$

Positive 15: no flip.

You can go for **at most 6 months**.

**Equation:** Plan A costs \$20 plus \$5 per GB. Plan B costs \$35 plus \$2 per GB. When do they cost **the same**?

$g$ = number of GB

$$\begin{aligned} 20 + 5g &= 35 + 2g \\ \underline{-2g} & \quad \underline{-2g} \\ 20 + 3g &= 35 \\ \underline{-20} & \quad \underline{-20} \\ 3g &= 15 \\ \frac{3g}{3} &= \frac{15}{3} \\ g &= 5 \end{aligned}$$

Same cost at **5 GB**. Check: $20 + 25 = 45$ and $35 + 10 = 45$ ✓